Class 3, Week 10: Can I trust this model? Overfitting, cross-validation and leakage

Lecturer guide for CML3112, CML3113 and CML3111. Keep it private.

The class in one paragraph

How do we know a model that predicts cooling load will still work on a building it has never seen?

Welcome back. In the first two classes you built models that predict cooling load and models that answer yes or no. Today we ask the question an engineer must always ask: can I trust this? A score that looks good is not enough. We learn how a model fools us, and how to test it so that it cannot.

When each cohort meets

Cohort	Programmes	Date	Time and room
CML3112	Civil Engineering and Architecture	Tuesday 17 November 2026	11:00 to 14:00, Room 106
CML3113	Mining and Petroleum Engineering	Friday 20 November 2026	08:00 to 11:00, Room 105
CML3111	Electrical, Mechatronics, Robotics and Communications	Friday 20 November 2026	11:00 to 14:00, Room 115

Before the class

- Before the class, run the notebook in real Colab and compare the depth-10 numbers for random_state 1, 17 and 99 with the row "Depth-10 tree, random_state 1, 17, 99" of the table below (correct for every cohort; real scikit-learn 1.9.1 gives exactly these). Another version may differ in the second decimal.
- The slide map below shows the CML3112 titles. Titles carry no cohort numbers, but the numbers on the slides differ by cohort: use the per-cohort table.
- Decision on choosing max_depth (tuning a hyperparameter): the simplest depth within 5 percent of the lowest validation MAE. The slides say plainly where the curve is flat, where the chosen tree also overfits its training rows, and that the chosen depth moves with the seed (CML3111 and CML3112: 6, 7, 8 for seeds 1, 17, 99; CML3113: 5 for all three).
- Cross-validation spread is the sample standard deviation (divide by k - 1, np.std(scores, ddof=1)), as in the project Task 5. The Class 2 StandardScaler divides by n: say once that these are two different jobs.
- Open the deck in Slide Show mode and the lab notebook in Colab before anyone arrives. Run the notebook from the top once so you know it works on your account.
- Put the lab notebook and engineering_data.csv in the shared class folder and post the link. Learners need both files in Colab, in the Files panel.
- Write the "check your numbers" table on the board before the lab (numbers for your cohort are in the table below).
- Say early that the test group E6 stays sealed. It is opened once, after the model and any threshold are chosen; the chosen model is then refitted on E1 to E5 and scored once on E6 (Class 4 shows this). Say that the validation fold in K-fold cross-validation is a different thing from the test set E6.
- The deck has seven formula slides (MAE and RMSE, training and validation error, the generalisation gap, the simplest-model-within-a-tolerance rule, K-fold mean and standard deviation, GroupKFold, the pipeline rule). Each names every symbol and works the cohort numbers. Students can revise from the PDF alone.
- All three cohorts use the same deck and the same clock for this class.

The clock

Tuesday cohort (CML3112): the first full teaching of this class. Friday cohorts (CML3113, CML3111): the same deck and the same clock as the Tuesday cohort. There is no separate recap version for this class.

CML3112, Tuesday

Starts	Minutes	What happens
0:00	10	Recall from Class 2, and why trust matters
0:10	40	Concepts, part 1: overfitting, validation curve, cross-validation
0:50	30	Concepts, part 2: data leakage, random seed, pipeline
1:20	15	Break
1:35	75	Lab: validation curve, cross-validation, leakage, seeds
2:50	10	Close and how this class counts

CML3113 and CML3111, Friday

Starts	Minutes	What happens
0:00	10	Recall from Class 2, and why trust matters
0:10	40	Concepts, part 1: overfitting, validation curve, cross-validation
0:50	30	Concepts, part 2: data leakage, random seed, pipeline
1:20	15	Break
1:35	75	Lab: validation curve, cross-validation, leakage, seeds
2:50	10	Close and how this class counts

Slide map

The deck has two halves. Slides up to the divider are concepts: no laptops. After the divider come the lab slides and the notebook. You switch once. The titles below are from the CML3112 deck. The other two decks have the same slides in the same order; where a title carries a number, their own number replaces it (see the per-cohort table).

Slide	Title	Part	Friday recap
1	Can I trust this model? Overfitting, cross-validation and leakage	Concepts	
2	Today in one view: what you will leave with	Concepts	
3	Your three hours: ideas first, then the lab	Concepts	
4	Three questions from last time	Concepts	
5	Answers to the three questions from Class 2	Concepts	
6	Words we use today	Concepts	
7	A model that looks good on paper can fail on a new machine or site	Concepts	
8	A model must do well on rows it has never seen	Concepts	
9	Underfitting, a good fit and overfitting look like three students	Concepts	
10	Reminder from Class 1: MAE and RMSE measure the prediction error	Concepts	
11	The validation curve: as max_depth grows, training error keeps falling but validation error stops improving	Concepts	
12	Training error, validation error and the generalisation gap are short formulas	Concepts	
13	The generalisation gap is the validation error minus the training error	Concepts	
14	Five depths in one table: leaf nodes, errors and the generalisation gap	Concepts	

Slide	Title	Part	Friday recap
15	We choose max_depth on the validation set and keep the test set sealed	Concepts	
16	Tuning a hyperparameter: choose the simplest max_depth within 5 percent of the lowest validation error	Concepts	
17	One split can be lucky, so K-fold cross-validation lets every fold take a turn as the validation fold	Concepts	
18	K-fold gives k scores, and we report their mean and their standard deviation	Concepts	
19	By hand with four illustrative scores: the mean is 3 and the standard deviation is 0.82	Concepts	
20	The same steps on our five real fold scores give their mean and standard deviation	Concepts	
21	GroupKFold holds out a whole equipment group in each fold, which is how the model is really used	Concepts	
22	GroupKFold uses the same formulas, with equipment groups as the folds	Concepts	
23	The same max_depth 4 tree gets three different scores, depending on how we test it	Concepts	
24	Data leakage is information the model would not have at prediction time reaching it during training	Concepts	
25	Data leakage comes in three common kinds	Concepts	
26	Train-test contamination: validation rows in the training set make the score look perfect	Concepts	
27	A Pipeline fits the imputer on the training set only, which prevents preprocessing leakage	Concepts	
28	The pipeline rule: fit on the training set only, then apply the same numbers	Concepts	
29	Target leakage: a feature built from the target gives a great score that cannot be repeated	Concepts	
30	Keep this checklist: five questions before you trust a score	Concepts	
31	The same code can give different numbers, so we fix the random seed (random_state) and write it down	Concepts	
32	Write down what you ran, so anyone can repeat it	Concepts	
33	Trust comes from four habits	Concepts	
34	Read the workflow before you write it	Concepts	
35	Part 2: the lab	Lab	
36	The lab has six steps, and the first five end with a check	Lab	
37	Check your numbers against these before you move on	Lab	
38	Four problems you will meet, what they mean and what to do	Lab	
39	How this class counts towards your coursework	Lab	
40	Save the notebook and keep the checklist	Lab	

Questions to ask, and a good answer

Ask	A good answer
Why does the training error keep falling as the tree gets deeper?	A deeper tree has more leaf nodes, so it can keep an answer for almost every row. That is overfitting.
How do we choose max_depth, and which rows do we use?	Tuning a hyperparameter: the simplest depth whose validation MAE is within 5 percent of the lowest (parsimony). Validation set only; the test set stays sealed.
Why does the linear regression model beat every tree here?	The target rises smoothly with load and ambient temperature, close to a straight line. A tree can only make steps.
Why divide by k - 1 for the standard deviation of fold scores?	It is the sample standard deviation, the usual choice for a few scores and the one the project uses: np.std(scores, ddof=1).
Why is the validation set a weak guide on its own?	It has only 15 rows, so its scores are noisy.
What does the standard deviation of the fold scores tell you?	How much the score depends on which rows were held out. A big standard deviation means less trust in the mean.
Why use GroupKFold?	Plain K-fold puts rows of all four groups in every fold, so the model has met each machine. GroupKFold tests on a group it has never met, as in real use.
What is the bias-variance trade-off?	A model that is too simple misses the pattern (high bias, underfitting). One that is too flexible chases noise (high variance, overfitting). We look for the middle.
Name the three kinds of data leakage.	Train-test contamination (validation rows in the training set), preprocessing leakage (an imputer or scaler fitted on all rows), target leakage (a feature built from the target).
Why is a leaky score of zero not good news?	It means the model saw the answers. The honest score is the one to quote.
Why do we write down the random seed (random_state)?	So that someone else, or you later, gets the same numbers from the same code.

Ideas learners often get wrong

What a learner may say	What to say back
The model with the lowest training error is the best model	Training error can reach zero by overfitting (fitting the noise in the training set). Judge on held-out rows.
More folds in cross-validation always give a better model	Folds give a better estimate of the score. They do not change the model itself.
Data leakage always shows as an error message	It shows as a score that is too good. Only the checklist and the order of steps prevent it.
If the score did not change, there was no leakage	Preprocessing leakage did not move the score here because only one value is missing. It is still a bad habit.
A random seed is a trick to get a better number	A seed only makes a random choice repeat. Trying seeds until one looks good is its own form of cheating.
The chosen max_depth is a fact about the data	With only 15 validation rows it can move with the seed (6, 7 or 8 for two cohorts). Cross-validation gives a steadier picture.
The chosen depth is a good fit, so its gap must be small	We choose by validation error. Where the curve is flat, the chosen tree can still overfit its training rows (CML3111 depth 7 has a gap of 3.79 deg C).

Walking the lab

Step	What to look for
Step 1: prepare	Rows left to model with: 88. Training, validation, test rows: 58, 15, 15. Anyone with different numbers ran a cleaning cell twice.
Step 2: validation curve	Ten lines of training MAE, validation MAE and generalisation gap, then each depth to 3 decimals with passes or fails. The chosen max_depth is printed. Ask why we accept the simplest depth within 5 percent of the lowest validation MAE.
Step 3: cross-validation	Five fold scores with the groups in each validation fold (all four every time), a mean and a sample standard deviation (ddof=1), then four group scores, sorted E1 to E4 (scikit-learn holds them out in its own order). The shortcut with cross_val_score must give the same means.
Step 4: data leakage	Honest and leaky MAE at depth 10. The leaky one is 0.00 or 0.01. The preprocessing leakage demo prints two different medians and the same MAE. The target-leakage feature gives a very small error.
Step 5: seeds and record	Three seeds, three different numbers, and the chosen max_depth for each seed. The run record prints versions: they will differ from the stand-in used to check the notebook, which is fine.
Step 6: conclusion and save	Three written sentences, then restart and run all, then save a copy.

The numbers each cohort should see

Check	CML3112 (kW)	CML3113 (kWh/t)	CML3111 (deg C)
Training, validation, test rows	58, 15, 15	58, 15, 15	58, 15, 15
Depth 1, validation MAE	5.30 kW	1.56 kWh/t	7.86 deg C
Depth 10, training and validation MAE	0.00 and 2.77	0.00 and 1.18	0.00 and 3.95
Chosen max_depth (simplest within 5 percent of the lowest)	7	5	7
Chosen max_depth with random_state 1, 17, 99	6, 7, 8	5, 5, 5	6, 7, 8
Depth-4 tree on E5 (one split), MAE	3.39	1.20	4.79
K-fold, k = 5: mean and sample standard deviation (ddof=1)	3.27 and 0.33	1.16 and 0.26	4.45 and 0.37
GroupKFold, k = 4: mean and sample standard deviation (ddof=1)	2.85 and 0.52	1.09 and 0.17	4.15 and 0.65
Honest and leaky MAE, depth 10	2.77 and 0.00	1.18 and 0.01	3.95 and 0.00
ambient_c median: training rows, and training plus validation rows	26.07 and 25.37	26.07 and 25.37	26.07 and 25.37
Target-leakage feature, max_depth 4, MAE	0.54	0.31	1.09
Depth-10 tree, random_state 1, 17, 99	2.93, 2.77, 3.02	1.21, 1.18, 1.28	4.70, 3.95, 3.85

If things go wrong

- If the concepts run over, skip the by-hand table with the illustrative scores and say: the standard deviation is the typical distance of a fold score from the mean. You can do the arithmetic in the lab.
- If the concepts still run over, skip the GroupKFold formula slide: it repeats the K-fold formulas with groups as the folds.

- If only thirty minutes of lab remain, do steps 1 to 3 and the first kind of leakage in step 4. Set the rest as practice for next time.
- If wifi fails in the room, the notebook runs unchanged in VS Code with the same libraries.
- If several learners are stuck at the same cell, stop the room, run that cell on the projector and let them copy the fix.

After the class

- Share the lab notebook and engineering_data.csv again so absent learners can follow.
- Tell the class that nothing is submitted today, and that the lecturer confirms assignment instructions and deadlines in writing.
- Note any question you could not answer and start the next class with it.

Speaker script

Every slide carries its spoken script in the PowerPoint notes (open Presenter View). The scripts are first person and are meant to be read or paraphrased. Pauses are marked by questions.